Development of a Rigorous Modeling Framework for Solvent-Based CO_2 Capture. Part 2: Steady-State Validation and Uncertainty Quantification with Pilot Plant Data

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Supporting Information for Publication

Table S1. Summary of NCCC steady state data (absorber variables)

Case No.	Lean Solvent Flowrate (kg/hr)	Lean Solvent Temperature (°C)	Lean Solvent CO_2 Loading (mol CO_2/MEA)	Nominal Lean Solvent MEA Weight Fraction	Gas Flowrate (kg/hr)	Gas Temperature (°C)	CO_2 mol%	Column Pressure (kPa)	Number of Beds (Intercoolers)
K1	6804	40.97	0.145	0.298	2248	42.48	11.55	108.82	3 (2)
K2	11794	40.52	0.247	0.312	2243	44.94	11.4	107.06	3 (2)
K3	3175	45.68	0.091	0.295	2242	43.67	10.58	107.78	3 (2)
K4	3175	46.72	0.083	0.310	2243	44.73	11.45	107.65	3 (2)
K5	6804	41.57	0.108	0.306	2235	43.78	9.19	106.94	3 (2)
K6	6804	40.87	0.347	0.307	2237	42.18	9.12	107.10	3 (2)
K7	11791	40.62	0.399	0.288	2233	44.72	9.18	107.30	3 (2)
K8	11643	40.57	0.154	0.285	2237	42.47	9.2	107.26	3 (2)
K9	3175	42.66	0.239	0.311	2232	44.87	9.22	107.49	3 (2)
K10	3175	48.59	0.062	0.297	2237	42.15	9.18	107.71	3 (2)
K11	6804	40.83	0.161	0.293	2237	44.40	10.01	107.20	3 (2)
K12	6804	40.92	0.160	0.293	2231	43.24	8.24	106.37	3 (2)
K13	6804	41.93	0.164	0.303	2238	42.09	9.35	107.56	3 (0)
K14	8845	40.41	0.224	0.303	2895	47.67	9.01	107.86	3 (2)
K15	8845	40.60	0.224	0.313	2908	42.83	9.07	108.08	3 (2)
K16	4128	44.45	0.124	0.329	2903	45.42	9.21	108.70	3 (2)
K17	6804	42.03	0.168	0.307	2240	41.03	9.19	108.12	2 (0)
K18	6804	42.24	0.141	0.302	2271	46.07	10.19	109.18	1 (0)
K19	11793	40.90	0.184	0.278	1440	46.18	11	108.02	1 (0)
K20	3175	45.30	0.075	0.276	1324	46.09	10.98	108.12	1 (0)
K21	3175	45.18	0.074	0.271	1366	46.11	10.18	107.76	2 (0)
K22	6804	41.24	0.130	0.281	1379	46.06	10.92	107.51	2 (1)
K23	6804	41.23	0.138	0.281	1418	46.11	10.13	107.37	2 (1)

Table S2. Summary of NCCC steady state data (stripper variables)

Case No.	Rich Solvent Flowrate (kg/hr)	Inlet Solvent Temperature (°C)	Outlet Solvent Temperature (°C)	Rich Loading (mol CO_2/MEA)	Rich MEA Weight fraction	Operating Pressure (kPa)	Reboiler Duty (kW)
K1	7242	104.8	120.18	0.384	0.300	183.87	431
K2	12284	104.8	117.43	0.385	0.314	182.06	430
K3	3335	97.4	122.15	0.475	0.313	184.36	427
K4	3343	97.6	122.53	0.470	0.328	184.15	427
K5	7212	109.0	121.68	0.295	0.308	183.43	673
K6	7063	95.7	110.21	0.469	0.309	179.88	173
K7	12092	93.2	103.76	0.471	0.289	184.41	170
K8	12043	110.1	120.35	0.275	0.289	183.45	677
K9	3337	98.4	117.69	0.474	0.318	182.84	166
K10	3358	94.5	124.00	0.477	0.309	182.67	671
K11	7241	107.4	119.88	0.378	0.294	182.67	425
K12	7173	109.2	120.56	0.341	0.294	183.81	422
K13	7078	107.6	120.65	0.360	0.307	183.19	419
K14	9393	104.8	119.35	0.420	0.302	184.02	420
K15	9349	103.5	119.32	0.413	0.313	182.00	420
K16	4347	98.6	122.94	0.476	0.349	181.86	419
K17	7051	107.7	120.49	0.354	0.314	181.94	418
K18	7099	105.9	120.38	0.349	0.304	184.06	425
K19	12161	107.9	118.00	0.276	0.281	183.76	427
K20	3369	99.7	120.39	0.393	0.279	183.56	438
K21	3370	99.7	120.36	0.385	0.272	184.36	437
K22	7161	108.3	119.16	0.291	0.271	183.13	427
K23	7146	108.4	119.32	0.283	0.282	181.80	427

Table S3. Comparison of model predictions of CO_2 capture efficiency with data values calculated from the liquid and gas side mass balances.

	Experimental Data		Model Predictions	
Case No.	Liquid Side	Gas Side	Original Model	Model with Composition Uncertainty
K1	89.45	99.91	99.96	99.95
K2	93.26	99.49	99.84	99.78
K3	72.90	83.57	77.33	76.46
K4	70.86	78.10	76.50	75.72
K5	90.69	99.53	99.98	99.98
K6	58.89	59.03	68.16	60.59
K7	57.50	54.76	69.82	57.67
K8	93.92	98.07	99.95	99.95
K9	52.30	55.48	57.55	54.85
K10	89.69	98.43	95.35	94.69
K11	91.63	99.75	99.93	99.92
K12	92.47	99.61	99.93	99.92
K13	89.84	97.98	98.36	98.11
K14	92.89	98.27	99.32	98.88
K15	91.72	99.42	99.44	99.11
K16	84.74	93.54	87.54	86.06
K17	88.92	97.61	98.18	97.88
K18	85.72	92.85	94.36	93.71
K19	93.18	98.21	98.91	98.73
K20	89.92	95.55	98.02	97.81
K21	90.05	96.32	99.72	99.69
K22	94.58	99.49	99.97	99.97
K23	92.61	99.58	99.96	99.96

Table S4. Comparison of model predictions of CO_2 loading in the outlet lean solvent from the regenerator

Case No.	Original		With Composition Discrepancy	
	Data	Model	Data	Model
K1	0.145	0.169	0.152	0.170
K2	0.247	0.273	0.260	0.279
K3	0.091	0.086	0.096	0.086
K4	0.083	0.090	0.087	0.090
K5	0.108	0.116	0.113	0.116
K6	0.347	0.371	0.365	0.372
K7	0.399	0.425	0.419	0.427
K8	0.154	0.165	0.162	0.166
K9	0.239	0.243	0.251	0.243
K10	0.062	0.059	0.065	0.059
K11	0.161	0.166	0.169	0.167
K12	0.160	0.164	0.168	0.165
K13	0.164	0.173	0.172	0.174
K14	0.224	0.240	0.235	0.243
K15	0.224	0.244	0.235	0.247
K16	0.124	0.119	0.13	0.119
K17	0.168	0.173	0.177	0.174
K18	0.141	0.168	0.148	0.169
K19	0.184	0.215	0.193	0.219
K20	0.075	0.084	0.079	0.084
K21	0.074	0.078	0.078	0.078
K22	0.130	0.153	0.137	0.153
K23	0.138	0.156	0.145	0.156

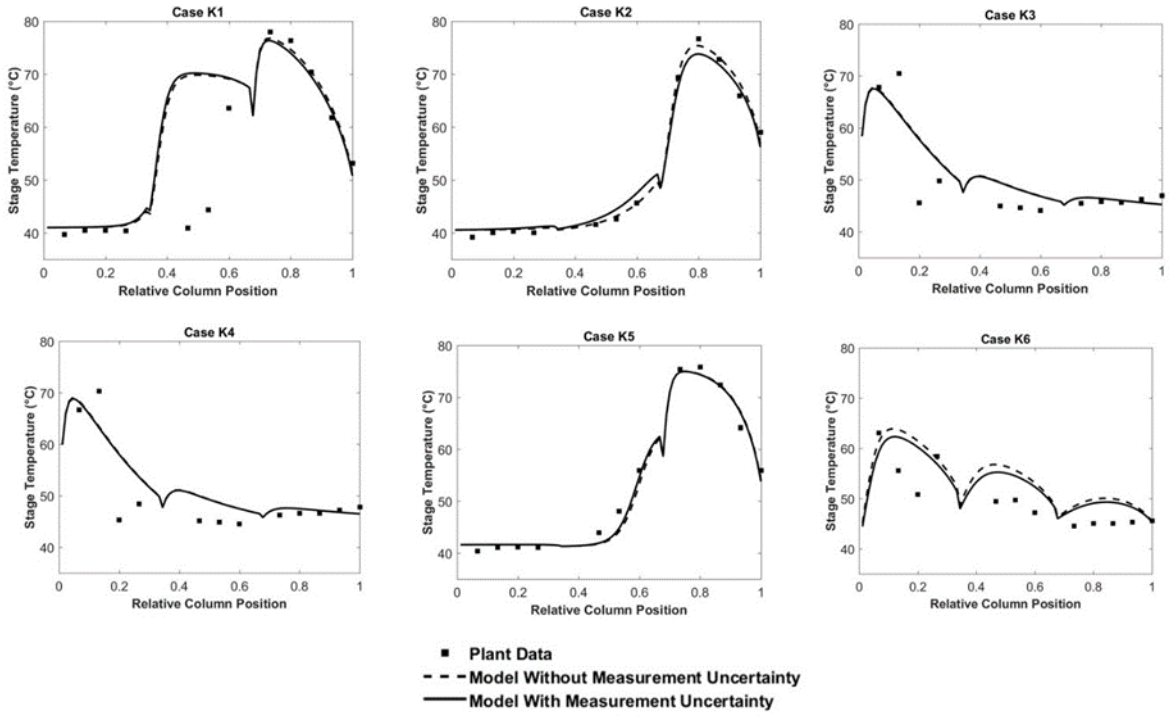


Figure S1: Comparison of model and data absorber temperature profiles for Cases K1-K6.

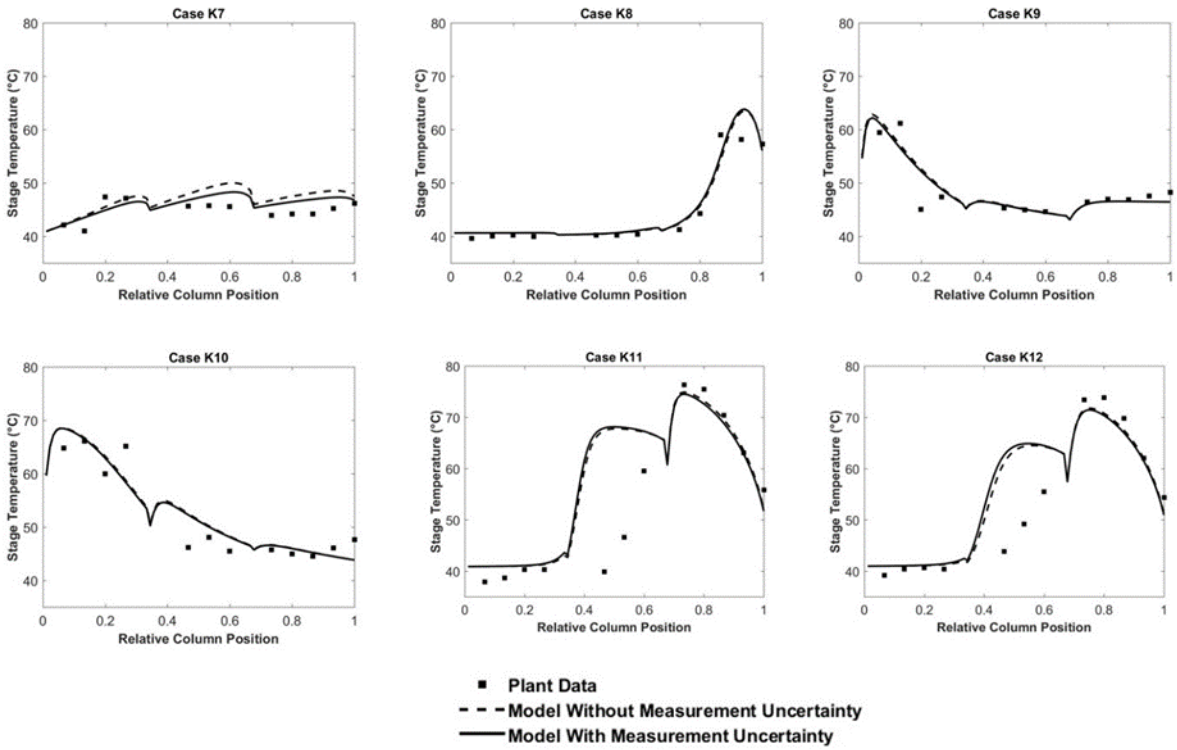


Figure S2: Comparison of model and data absorber temperature profiles for Cases K7-K12.

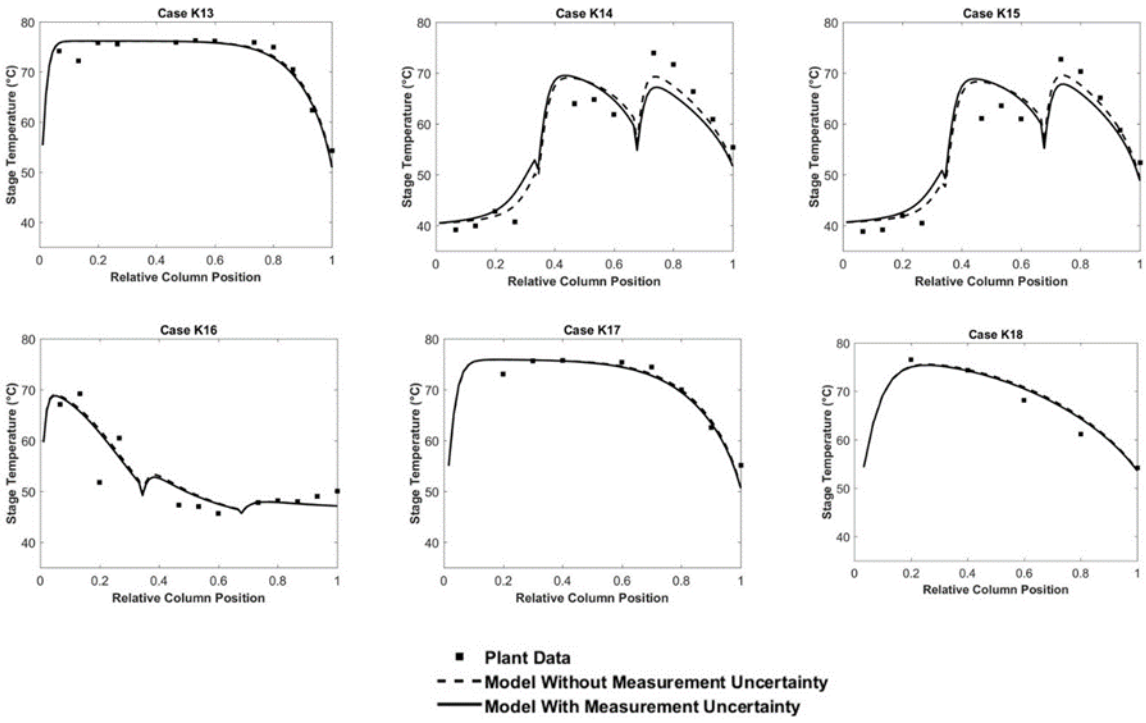


Figure S3: Comparison of model and data absorber temperature profiles for Cases K13-K18.

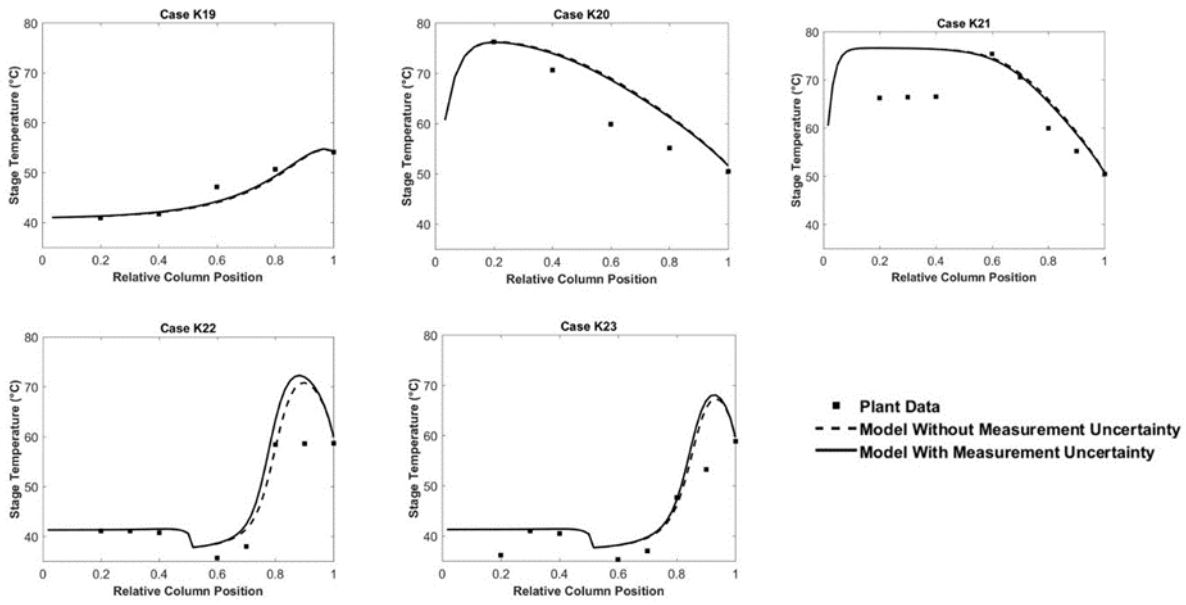


Figure S4: Comparison of model and data absorber temperature profiles for Cases K19-K23.

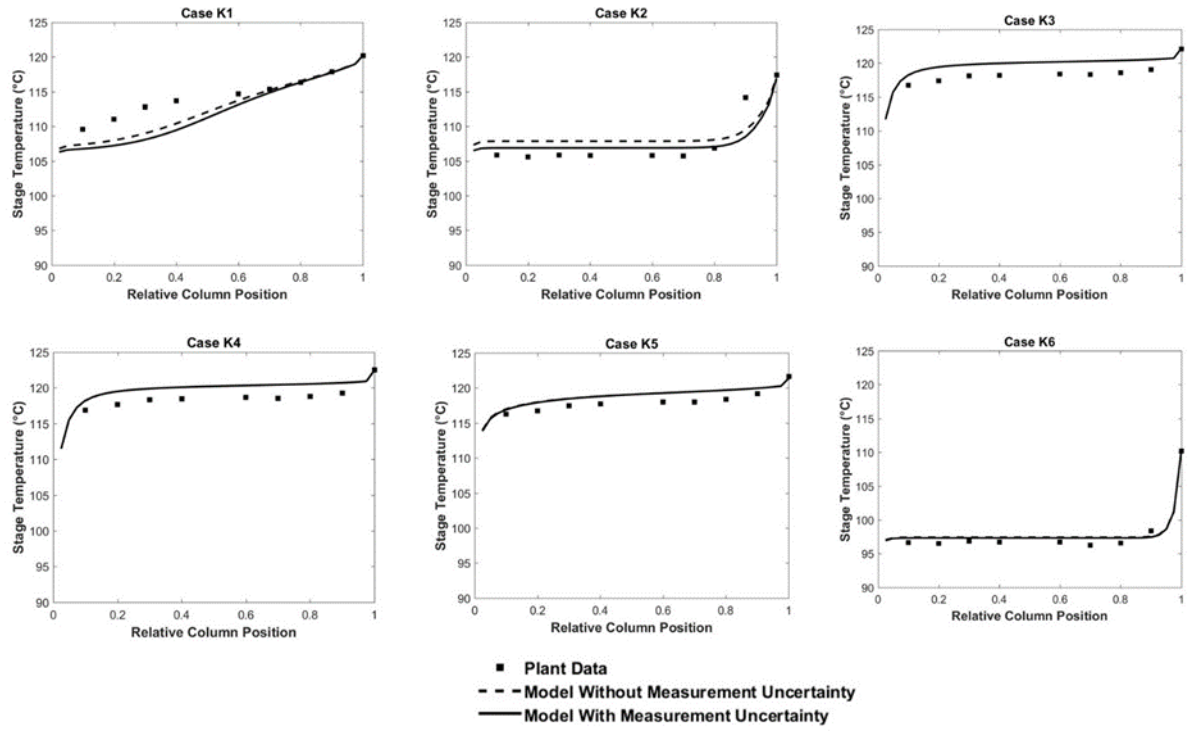


Figure S5: Comparison of model and data absorber temperature profiles for Cases K1-K6.

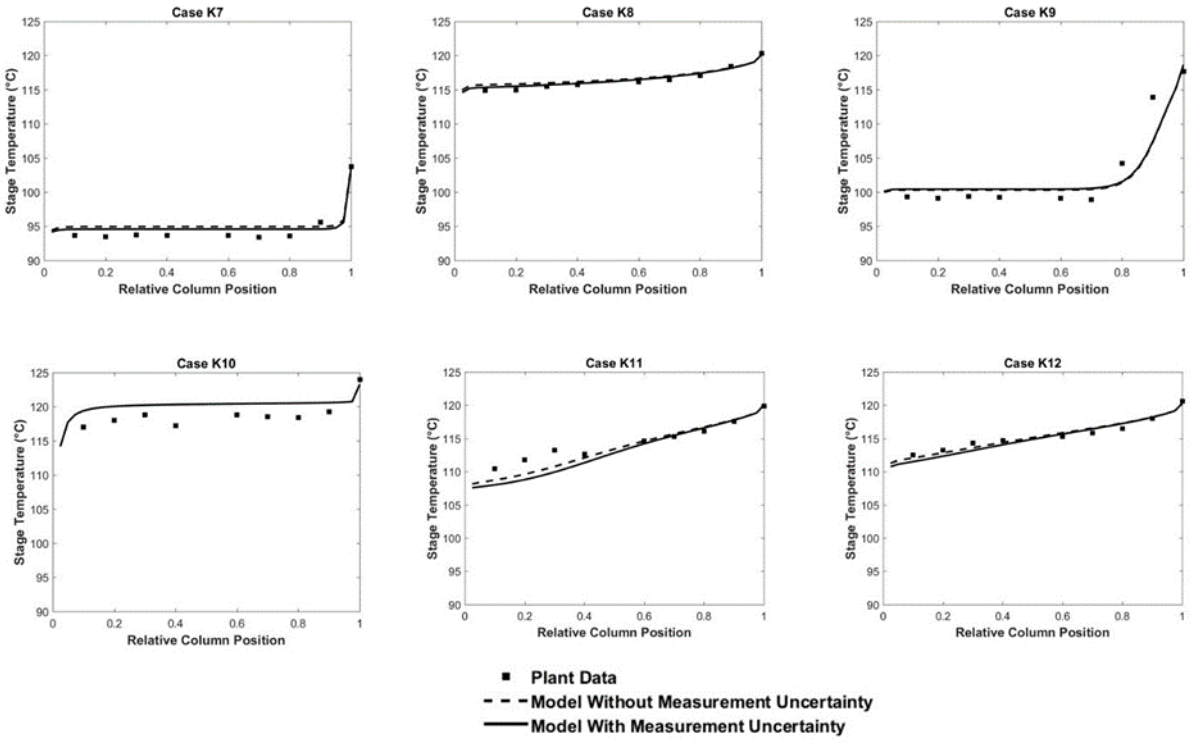


Figure S6: Comparison of model and data absorber temperature profiles for Cases K7-K12.

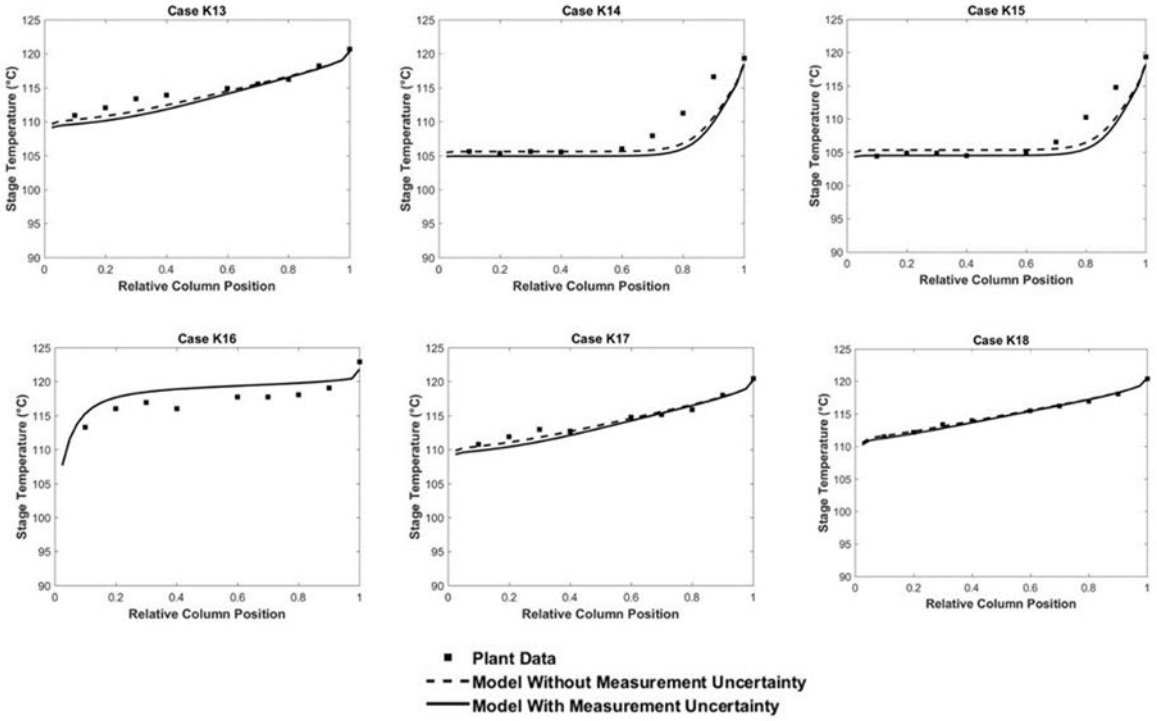


Figure S7: Comparison of model and data absorber temperature profiles for Cases K7-K12.

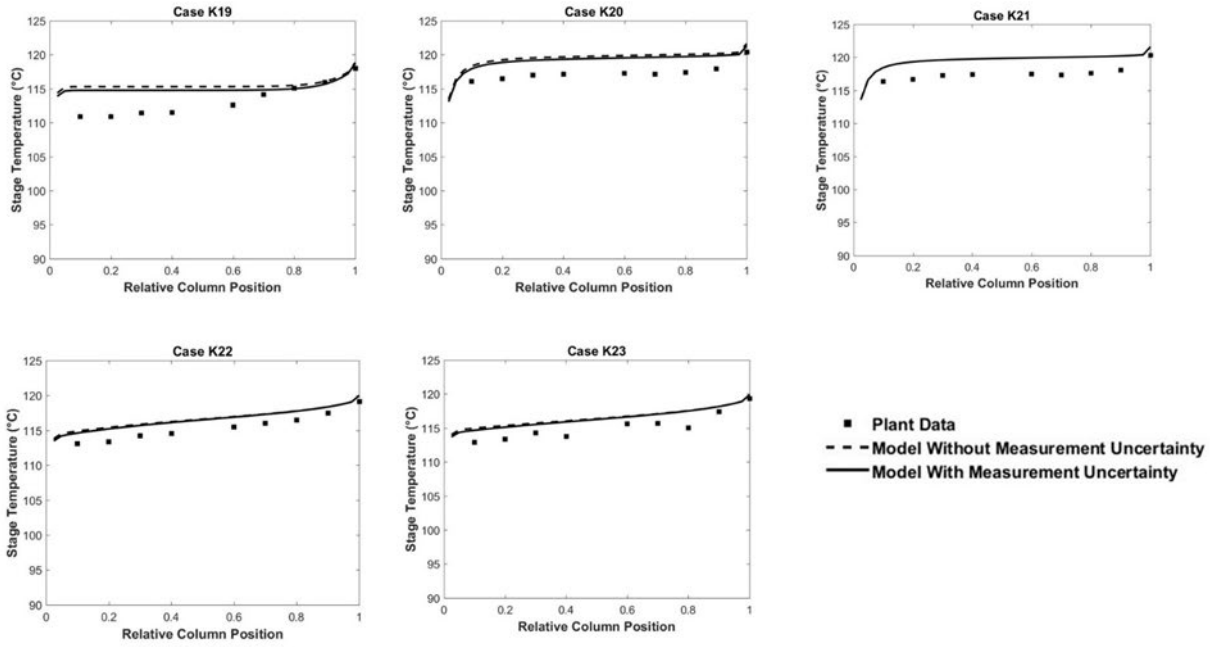


Figure S8: Comparison of model and data absorber temperature profiles for Cases K7-K12.